

Traffic control table

- Routing table: Contains information needed for traffic routing table.
- Every network device capable of IP communication has a routing with board
- Routing based on network address
- The network device matches the destination IP address in the header of the incoming IP packet. using the routing table information, it forwards the appropriate interface.

Target network	Gateway	Mask	Metrics	Interface
10.1.1.0	0.0.0.0	255.255.255.0 2		eth0
10.1.2.0	10.1.2.1	255.255.255.0 3		eth1
0.0.0.0	10.1.1.1	0.0.0.0	1	eth0

Traffic controllers (basic) operation

1. The router receives the packet arriving on the input interface.
2. The router matches the packet's destination address to the routing table rows. If the destination address matches multiple rows, the row with the longest prefix is considered to be the match.
3. If there is no matching row, the destination is unreachable, the packet cannot be forwarded. The router drops the packet and sends an ICMP error message to the sender.
4. If a matching queue exists, the packet is forwarded on the output interface in it (with data link layer encapsulation) to the neighbor specified as the next hop, or to the destination if there are no more hops.

IP address matching

1. Sort the rows of the routing table in descending order by prefix length . $N=1$. This ensures that in the event of multiple matching rows, the one with the longest prefix will be returned.
2. If row N does not exist in the table, then there is no matching row, and
It's over.
3. Bitwise AND between the packet's destination address and the network mask of line N .
we are performing an operation.
4. If the result of the bitwise AND operation is the same as the target network value of row N , then the address matches row N and ends.
5. $N=N+1$; continue at point 2.

Routing table query

route -n

netstat -nr

ip route show

Adding/deleting a direct access network

```
# route add -net 1.2.0.0/16 eth0
```

South

```
# ip route add 1.2.0.0/16 dev eth0
```

South

Adding/deleting a non-direct access network

```
# route add -net 2.3.0.0/16 gw 1.2.3.1 eth0
```

South

```
# ip route add 2.3.0.0/16 via 1.2.3.1 dev eth0 del
```

Add/Delete Default Gateway

```
# route add default gw 10.1.1.1
```

```
# route del default
```

Allow through traffic

```
# sysctl -w net.ipv4.ip_forward=1
```


Task

- In the classroom, create 4 /26 subnets from the initial network 10.0.1.0/24.
The student machines belong to one IP network per column, everyone should configure a free address on the *eth0* interface from the appropriate subnet.
- Ping another device in our column (our own network) node.
- Ping another node in the column. Do we get a response?
Let's explain the phenomenon!
- Modify our routing table to include all subnets in the room.
we can reach it!